With Little Power Comes Great Responsibility

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Abstract

Despite its importance to experimental design, statistical power (the probability that, given a real effect, an experiment will reject the null hypothesis) has largely been ignored by the NLP community. Underpowered experiments make it more difficult to discern the difference between statistical noise and meaningful model improvements, and increase the chances of exaggerated findings. By meta-analyzing a set of existing NLP papers and datasets, we characterize typical power for a variety of settings and conclude that underpowered experiments are common in the NLP literature. In particular, for several tasks in the popular GLUE benchmark, small test sets mean that most attempted comparisons to state of the art models will not be adequately powered. Similarly, based on reasonable assumptions, we find that the most typical experimental design for human rating studies will be underpowered to detect small model differences, of the sort that are frequently studied. For machine translation, we find that typical test sets of 2000 sentences have approximately 75% power to detect differences of 1 BLEU point. To improve the situation going forward, we give an overview of best practices for power analysis in NLP and release a series of notebooks to assist with future power analyses.[1]

1 Introduction

Despite its importance to empirical evaluation, relatively little attention has been paid to statistical power in NLP. In particular, *if it is the case that typical experiments in NLP are underpowered*, not only would we expect many meaningful improvements to go undetected, we would also expect many apparently significant differences to be exaggerated (Gelman and Carlin, 2014). In this paper, we build on past work calling for greater rigor in evaluation (McCoy et al., 2019; Azer et al., 2020), including the need for careful hypothesis testing (Koehn, 2004; Berg-Kirkpatrick et al., 2012; Søgaard et al., 2014; Dror et al., 2018), and show why and how power matters to NLP, addressing challenges unique to this domain.

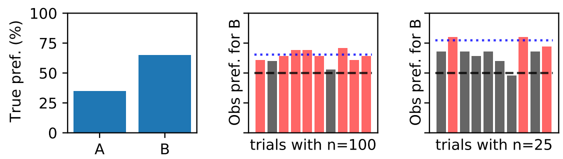

Figure 1: Cartoon example of statistical power in comparing two models: 65% of all people in the population always prefer system B (left). A comparison using a sample of 100 people would be well-powered (middle): over 80% of such samples will show a significant difference (plotted in red) from the null hypothesis that the models are equally good (dashed line). In samples of 25 people (right), far fewer tests will be significant (power $\approx$ 30%). Note that the observed mean of significant findings (dotted line) slightly overestimates the true proportion that prefer system B when $n = 100$ and more severely overestimates it when $n = 25$.

Roughly speaking, power is the probability that a statistical test will successfully detect a *true effect*. As an illustrative example, imagine comparing two dialog systems (see Figure 1). We want to know if people tend to prefer one system over the other. To test this, we will need multiple people to evaluate the systems. But how many? Once we have collected data, a statistical *test* will tell us if we can reject the null hypothesis the systems are equally good. Assuming the systems are not identical, statistical *power* is the probability that the experiment will return a significant result (or equivalently, it is one minus the probability of failing to detect the difference as significant). Although we don't know the magnitude of this difference, *power analysis* helps to estimate how much power an experiment

[1]https://github.com/dallascard/NLP-power-analysis

will have under various assumptions.

Power depends on multiple factors, including the statistical test used, the significance threshold, true effect size, variance, and sample size. All else being equal, experiments with larger samples will have greater power than smaller samples, as shown in Figure 1. Similarly, larger effects and those with less variance are easier to detect, and therefore require fewer samples for equivalent power. Importantly, note that if we *do* find a significant difference, this does *not* imply that the experiment had high power.[2]

Proceeding with a test that is *underpowered* (i.e., too few subjects or items; often taken to mean less than 80% power; Cohen, 1962) means that one is less likely to be able to draw any useful statistical conclusion from the experiment, and has contributed, in part, to the replication crisis in other fields (Button et al., 2013; Szucs and Ioannidis, 2017; Ioannidis et al., 2017). Routinely running experiments with low statistical power undermines the scientific enterprise. Not only will true effects go undetected; when significant effects are found, they are likely to be noisier and have lower positive predictive value (Button et al., 2013).

Moreover, significant findings from underpowered experiments are more likely to exaggerate or reverse the true effect – so-called Type-M (magnitude) and Type-S (sign) errors, respectively (Gelman and Carlin, 2014). This problem can lead to systematic distortions in the literature if only significant findings are published, especially if these results are based on underpowered experiments (Scargle, 1999). The effect of Type-M error can be seen in Figure 1; significant differences are less likely to be found in smaller samples (right), but among those tests that are significant, the observed difference will tend to exaggerate the true difference (left) by more than a larger sample (middle). For further discussion of Type-M and Type-S errors, please refer to Appendix B.

Here, we investigate how these issues affect NLP. Although retrospective analysis of power involves challenges, we present evidence that underpowered experiments are widespread in NLP research. Among human evaluations, we find most experimental designs involve too few items and/or raters to detect small effects (§5). For comparing models in terms of accuracy, we find that some widely used benchmark datasets, including MRPC and SST-2, are now too small to be able to properly measure future progress against top performing models (§3). We also introduce a novel approach to power analysis for machine translation and characterize power in experiments testing for differences in BLEU (§4). Finally, a survey of recent papers reveals a general lack of statistical evaluation and a dearth of detailed reporting (§5.1).

To improve future practice, we suggest broader adoption of power analyses prior to evaluation, provide guidance on running power analyses in NLP, and release a series of notebooks for this purpose.

2 Power Analysis for NLP

Because most NLP tasks do not take the form of standard experiments in other sciences (Kraemer and Blasey, 2015; Westfall et al., 2014), it is nontrivial to run power analyses for many tasks of interest. While we cannot cover every scenario, we present here a generalizable, simulation-based approach to power analysis, along with three sample applications, which can be extended as necessary. Such an approach is modular, reusable, and transparent, and encourages planning of analyses in advance of data collection.

Every power analysis requires assumptions, and there is not likely to be a single correct approach. Rather, the point is to make one's assumptions explicit, and include enough detail so as to account for whatever is likely to be observed. By using reasonable assumptions, one can help to ensure that one's experiment is sufficiently well-powered, In the case of NLP, this means that one recruits enough subjects, collects enough ratings, or uses a large enough test set.

The general procedure we suggest for power analysis is described in detail in Figure 2. At a high level, the idea is to estimate power by running simulations. Recall that power is the probability of detecting a true effect, conditional on the experimental setting (effect size, variance, etc.) and significance threshold. Thus, if one can translate these assumptions into a process for generating simulated data, we can estimate power by generating many simulated datasets using assumed or estimated parameter values, running each sample through a significance test, and reporting the proportion that are found to be significant.

[2]Using the observed outcome from a single experiment to compute power falls into the trap of post-hoc power analysis and is not recommended. For additional background on statistical power, power analysis, null-hypothesis significance testing, and post-hoc analysis, please refer to Appendix A.

Define a generative process $G(n, e^*, \mathbf{h})$ parameterized by number of items, n, hypothesized effect e^* for the statistic of interest E, and other relevant parameters $\mathbf{h}$ (e.g., variance). Also choose a statistical test $T(\mathcal{D})$, which returns a p-value p when performed on data $\mathcal{D}$ sampled from $G(n, e^*, \mathbf{h})$. Finally, choose the size of the dataset to be sampled, n, significance threshold, α, and number of repetitions, r.

1. For i in range(r):
   - sample a dataset of size n, $\mathcal{D}_i \sim G(n, e^*, \mathbf{h})$
   - compute the effect of interest on this sample, $e_i = E(\mathcal{D}_i)$
   - also compute a p-value according to the test of interest: $p_i = T(D_i)$
2. Power $\approx \frac{1}{r} \sum(\mathbb{I}[p_i \leq \alpha] \cdot \mathbb{I}[\text{sign}(e_i) = \text{sign}(e^*)])$

Figure 2: An algorithm for power analysis by simulation. For the example of comparing two systems presented in Figure 1, e^* is the assumed overall proportion of people who prefer system B, relative to the null hypothesis, $p = 0.5$, $G(n, e^*, \mathbf{h})$ is simply $\text{Binomial}(n, 0.5 + e^*)$, while e_i is the observed proportion of people who prefer system B in sample i, again relative to 0.5. For extensions to estimate Type-M and Type-S error, see Appendix B.

The key to generalizing this approach is to begin with the end in mind. In particular, if one plans to test for a difference between models, one needs to choose the statistical test that will be used. That test will determine the level of detail required in the generative process for simulating data.

To return to the opening example of evaluating dialog systems, we want to test if people prefer one system over the other (Ai et al., 2007). If we ignore the nuances of human preference for now (but see §5 for a more nuanced approach), and simply assume that each person either prefers system A or system B, the only assumption we need to make for a power analysis in this setting is the proportion of people in the population who prefer system B. We can then simulate samples of n people (each of whom independently has the same probability of preferring system B) as a draw from a binomial distribution, and repeat this thousands of times.[3] For each sample, we then test whether the proportion of people who prefer system B is significantly different from 0.5. The estimated power of this experiment would thus be the proportion of simulated differences that are found to be significant.[4]

The most difficult part of power analyses is estimating the relevant quantities, such as the *true* proportion of people that prefer system B. Note, however, that one can always compute what power would be for a range of possible values, and indeed, this is the recommended procedure. For estimating the relevant parameters within an NLP context, we will primarily rely on data from the literature, measurements on validation data, and estimates from external datasets (see §3.2). However, where appropriate, pilot studies may also be informative.

In the remainder of this paper, we consider three scenarios of interest in depth, and assess the state of power in the NLP literature for each.

3 Comparing Models on Accuracy

It is common in NLP research to look for models which improve over state of the art (SOTA) on various benchmarks. However, an important but rarely asked question is, *can these benchmarks support the kinds of comparisons we want to make?* Many have emphasized the need for proper significance testing to avoid spurious findings, but if an experiment's test set is small, the minimum detectable effect (MDE) size may be large: only large improvements will yield sufficiently powered comparisons (i.e., $\geq 80\%$ power). If an experiment is badly underpowered, it cannot provide useful evidence that one model achieves slightly better performance than another for the underlying data distribution. Reliance on such evidence risks leading to over-confidence about the relative ranking of various models. As we show in §3.3, there is legitimate reason to be concerned about this in the case of certain widely used benchmarks.

3.1 Significance test for comparing classifiers

The standard statistical test for comparing classifiers on paired data is McNemar's test (Dietterich, 1998; Dror et al., 2018), which uses the numbers of items where the models disagree (i.e., the off-diagonal elements in Table 1).[5] McNemar's test assesses whether $\chi^2 = \frac{(p_{10}-p_{01})^2}{p_{10}+p_{01}}$ is significant, and if so, rejects the null hypothesis that the distributions are the same.

[3] We don't need to address variance in this scenario, as the variance of a binomial distribution is a function of its mean.

[4] More direct solutions are available for some settings, including this one (see Appendix E.5), but we describe it using the generic approach from Figure 2 for the purpose of illustration. For all cases examined in this paper, simulations take only minutes on a laptop.

[5] Unpaired data (i.e., if two models are evaluated on different data drawn from the same distribution) requires a different approach, such as using a binomial test. See Appendix E.5 for extended discussion.